

mechanical properties of vinylogous urethane vitrimer and its influencing factor

Vinylogous urethane vitrimers exhibit tunable mechanical properties—with Young's modulus ranging from kilopascals to gigapascals, tensile strength up to 114 MPa, and glass transition temperatures spanning over 150°C—that are systematically controlled by crosslinking density, free amine content, catalytic additives, polymer matrix selection, and network architecture features such as block copolymer structures and dual dynamic/static crosslinks.

Abstract

Vinylogous urethane (VU) vitrimers exhibit mechanical properties spanning a remarkably wide range, with Young's modulus from approximately 31 kPa in soft polyhydroxyurethane systems to 2.4 GPa in rigid bulk-polymerized networks, tensile strength from 0.18 MPa to over 114 MPa, and glass transition temperatures from -49°C to above 110°C . This variability is systematically controlled by several key factors: crosslinking density shows a positive linear correlation with both modulus and T_g ; free amine content determines stress relaxation rates, with excess amine formulations relaxing in seconds versus hours for amine-deficient networks; and catalytic additives modulate exchange kinetics, with Brønsted acids reducing relaxation times by 5-fold and solvate ionic liquids lowering activation energy to 24 kJ/mol. Network architecture modifications including block copolymer structures, mechanically interlocked motifs, and dual static/dynamic crosslinks provide additional pathways to enhance mechanical performance while maintaining reprocessability.

Reprocessing studies consistently demonstrate excellent property retention over multiple cycles, with most systems showing no significant degradation after 3-4 grinding and hot-pressing cycles at temperatures ranging from $100-180^{\circ}\text{C}$. The trade-off between fast stress relaxation for reprocessing and dimensional stability during service can be addressed through temperature-reversible amine masking strategies or incorporation of phase-separated glassy domains that reduce creep by 18-46%. Bio-based formulations using lignin, β -myrcene, castor oil, and starch achieve competitive mechanical properties, demonstrating the versatility of VU chemistry for sustainable material design.

Paper search

We performed a semantic search using the query "mechanical properties of vinylogous urethane vitrimer and its influencing factor" across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We retrieved the 500 papers most relevant to the query.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Vinylogous Urethane Vitrimer Focus:** Does this study investigate vinylogous urethane vitrimers as the primary material of interest?
- **Quantitative Mechanical Properties:** Does this study report quantitative mechanical properties data (e.g., tensile strength, modulus, elongation at break, etc.)?
- **Mechanical Property Influencing Factors:** Does this study examine factors that influence mechanical properties (e.g., crosslink density, temperature, molecular weight, curing conditions)?
- **Study Type and Quality:** Is this study either an original experimental research article or a systematic review/meta-analysis on vinylogous urethane vitrimers that has been peer-reviewed?

- **Standardized Testing Methods:** Does this study use standardized mechanical testing methods (e.g., tensile testing, DMA, compression testing, rheology)?
- **Material Chemistry Specificity:** Is this study focused on vinylogous urethane chemistry rather than solely on other types of vitrimers (e.g., disulfide-based, transesterification-based) without vinylogous urethane components?
- **Mechanical Characterization Inclusion:** Does this study include mechanical characterization rather than reporting only synthesis methods without mechanical property evaluation?
- **Publication Quality:** Is this study a peer-reviewed publication rather than a conference abstract, patent, or non-peer-reviewed publication?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Vitrimer Composition:**

Extract the specific composition and formulation details of the vinylogous urethane vitrimer studied, including:

- Base materials used (e.g., acetoacetate precursors, diamine crosslinkers)
- Molar ratios of components
- Any additives (catalysts, acids, bases, etc.)
- Whether bio-based or synthetic precursors were used
- Any special structural features (e.g., mechanically interlocked networks, chain extenders)

- **Mechanical Properties:**

Extract all mechanical properties measured for the vinylogous urethane vitrimers, including:

- Young's modulus (MPa)
- Tensile strength (MPa)
- Toughness (MJ/m³)
- Elongation at break (%)
- Stress relaxation behavior
- Damping capacity
- Viscoelastic properties
- Glass transition temperature (T_g) Include numerical values with units and any statistical measures (standard deviation, error bars) when provided.

- **Influencing Factors:**

Extract all factors that were studied for their influence on mechanical properties of vinylogous urethane vitrimers, including:

- Chemical composition variations (different crosslinkers, chain extenders, additives)
- Processing conditions (temperature, time, pressure)
- Network architecture modifications
- Curing conditions
- Environmental factors (humidity, temperature during testing)

- Any other variables that were systematically varied to study their effects

- **Property-Factor Relationships:**

Extract the specific relationships between influencing factors and mechanical properties for vinylogous urethane vitrimers, including:

- How each factor affects each mechanical property (increases, decreases, optimal range)
- Quantitative relationships when available (e.g., 'increasing X from Y to Z increased modulus by N%')
- Synergistic or antagonistic effects between multiple factors
- Mechanisms explaining why factors affect properties
- Any threshold effects or non-linear relationships

- **Testing Conditions:**

Extract the experimental conditions under which mechanical properties of vinylogous urethane vitrimers were measured, including:

- Testing temperature
- Testing method/standard used (ASTM, ISO, etc.)
- Sample preparation method
- Crosshead speed or strain rate
- Sample geometry and dimensions
- Environmental conditions (humidity, atmosphere)
- Any pre-conditioning of samples

- **Reprocessability Effects:**

Extract information about how reprocessing affects the mechanical properties of vinylogous urethane vitrimers, including:

- Number of reprocessing cycles studied
- Reprocessing conditions (temperature, pressure, time)
- Changes in mechanical properties after each cycle
- Retention percentage of original properties
- Any degradation mechanisms observed
- Recovery of properties after reprocessing

Characteristics of Included Studies

This systematic review encompasses 40 sources examining the mechanical properties of vinylogous urethane (VU) vitrimers and the factors influencing these properties. The studies span from 2015 to 2025, representing the development of this field since the initial introduction of VU vitrimers.

Study	Full text retrieved?	Study Type	Primary Focus
Wim Denissen et al., 2015	Yes	Primary study	Network synthesis and characterization
Hu Xu et al., 2022	No	Primary study	Epoxy vitrimers with multiple recycling routes

Study	Full text retrieved?	Study Type	Primary Focus
Jun Zhao et al., 2021	No	Primary study	Mechanically interlocked vitrimers
P. Haida et al., 2023	No	Primary study	Starch-reinforced vitrimer composites
Youwei Ma et al., 2025	No	Primary study	Elastomeric VU thermosets for strain sensors
Mohammad Farkhondehnia et al., 2023	Yes	Primary study	Bio-based polyhydroxyurethane vitrimers
Giuseppe Soavi et al., 2023	No	Primary study	Phenoxo resin-based VU networks
J. Shin et al., 2022	No	Primary study	Solvate ionic liquid-enhanced vitrimers
Jie Liu et al., 2022	No	Primary study	Pseudo-cyclic VU vitrimers
Farhad Asempour et al., 2024	No	Primary study	Polymyrcene elastomer shape memory
C. Du et al., 2025	No	Primary study	β -myrcene block copolymer elastomers
Farhad Asempour et al., 2023	Yes	Primary study	Myrcene-based rubber property modulation
Fengbiao Chen et al., 2020	No	Primary study	Fusion of bio-based vitrimers
Jacob S. A. Ishibashi et al., 2020	Yes	Primary study	Block copolymer dynamic cross-linking
Tyler S. Stukenbroeker et al., 2017	No	Primary study	PDMS quenchable vitrimers
Siraphat Weerathaworn et al., 2024	No	Primary study	Block copolymer architecture effects
Stéphanie Engelen et al., 2024	Yes	Primary study	Vinylogous urea-urethane hydrogen bonding
Meijuan Xie et al., 2024	No	Primary study	Polyisoprene elastomers with PS phases
Ruhan Yang et al., 2023	No	Primary study	Bio-based epoxy thermosets
Seongon Jang et al., 2023	Yes	Primary study	Vitrimer electrolytes with lithium salts
Lisa Sougrati et al., 2023	Yes	Primary study	Lignin-based aromatic vitrimers
W. Denissen et al., 2017	Yes	Primary study	Chemical control of viscoelastic properties
P. Haida et al., 2020	No	Primary study	Acid-mediated autocatalysis

Study	Full text retrieved?	Study Type	Primary Focus
Filip Van Lijsebetten et al., 2022	Yes	Primary study	Reversibly trapped amines for creep suppression
L. Bai et al., 2020	No	Primary study	Zn(II)-ligand bond enhanced composites
Taylor Wright et al., 2019	No	Primary study	Photoactivated healable copolymers
Jie Liu et al., 2022a	No	Primary study	Reaction-structure-property mapping
Yachen Wang et al., 2022	No	Primary study	PDMS/cellulose nanocrystal composites
Z. Qiao et al., 2022	No	Primary study	Polybenzoxazine/VU for carbon fiber recycling
Lisheng Wu et al., 2024	No	Primary study	Cyclic olefin vitrimers via ROMP
Filip Van Lijsebetten et al., 2023	No	Primary study	Masked primary amines for flow control
Patrick Schütz et al., 2025	No	Primary study	Au-doped BCP nanocomposites
I. Baere et al., 2018	No	Primary study	Fiber-reinforced polymer composites
Filip Van Lijsebetten et al., 2022a	Yes	Primary study	Controlled plastic flow via masked amines
Youwei Ma et al., 2023	No	Primary study	Closed-loop recycling of VU vitrimers
Filip Van Lijsebetten et al., 2024	Yes	Primary study	Thermoswitchable acid catalysis
Shalini J. Rukmani et al., 2024	Yes	Primary study	Vitrimer viscosity modeling
Yangyang Zhu et al., 2020	No	Primary study	Bio-based vitrimers from castor oil
Swagata Mondal et al., 2024	Yes	Primary study	Doubly crosslinked networks for creep resistance
W. Denissen et al., 2018	Yes	Primary study	Vinylogous urea fiber composites

The majority of studies (32 of 40) were accessed only through abstracts, while 8 studies had full text available. All included sources are primary research studies focusing on synthesis, characterization, and property optimization of VU vitrimers.

Mechanical Properties of Vinylogous Urethane Vitrimers

Overview of Reported Mechanical Properties

The mechanical properties reported across studies vary considerably depending on the base polymer matrix, crosslinking density, and intended application. The following table summarizes the quantitative mechanical property data extracted from the included studies.

Property	Study	Value	Conditions/Notes
Storage/Young's Modulus			
Storage modulus	Wim Denissen et al., 2015	≈ 2.4 GPa	Bulk polymerized network
Young's modulus	Hu Xu et al., 2022	>1700 MPa	Epoxy vitrimer
Young's modulus	Jun Zhao et al., 2021	18.5 ± 0.9 MPa	Mechanically interlocked vitrimer
Elastic modulus	P. Haida et al., 2023	2 GPa	Starch-reinforced composite
Young's modulus	Mohammad Farkhondehnia et al., 2023	31 ± 1.5 to 133 ± 5.7 kPa	PHU-based vitrimers
Glassy modulus	J. Shin et al., 2022	$>10^9$ Pa	SIL-incorporated vitrimer
Young's modulus	Lisa Sougrati et al., 2023	0.2 to 420 MPa	Lignin-based vitrimers
Young's modulus	W. Denissen et al., 2017	2.0 MPa	Elastomeric network
Storage modulus	W. Denissen et al., 2018	~ 2.2 GPa	Vinylogous urea vitrimer
Tensile Strength			
Tensile strength	Hu Xu et al., 2022	>70 MPa	Epoxy vitrimer
Tensile stress	C. Du et al., 2025	3.1 ± 0.6 MPa	Diblock copolymer vitrimer
Tensile strength	Farhad Asempour et al., 2023	0.18 ± 0.03 to 1.68 ± 0.02 MPa	Myrcene-based rubber
Tensile strength	Ruhan Yang et al., 2023	31.5 to 114.6 MPa	Bio-based epoxy thermosets
Tensile strength	Lisa Sougrati et al., 2023	0.2 to 17.2 MPa	Lignin-based vitrimers
Failure stress	W. Denissen et al., 2018	~ 64 MPa	Vinylogous urea composite
Toughness			
Toughness	Jun Zhao et al., 2021	3.7 ± 0.1 MJ/m ³	Mechanically interlocked
Impact strength	Z. Qiao et al., 2022	up to 16.9 kJ/m ²	Polybenzoxazine/VU system
Elongation at Break			
Elongation at break	W. Denissen et al., 2017	140%	Elastomeric network
Elongation at break	Farhad Asempour et al., 2023	30% to 83%	Varies with crosslink density
Elongation at break	C. Du et al., 2025	$\sim 90\%$	Block copolymer vitrimer
Elongation at break	Ruhan Yang et al., 2023	9.5 to 150.0%	Bio-based epoxy
Elongation at break	Lisa Sougrati et al., 2023	22 to 98%	Lignin-based vitrimers
Elongation at break	W. Denissen et al., 2018	$\sim 6.4\%$	Rigid vinylogous urea

Property	Study	Value	Conditions/Notes
Glass Transition Temperature			
Tg	Wim Denissen et al., 2015	>80°C	Bulk polymerized
Tg	Hu Xu et al., 2022	>75°C	Epoxy vitrimer
Tg	Farhad Asempour et al., 2024	~45°C	Polymyrcene elastomer
Tg	Farhad Asempour et al., 2023	−49 to −5°C	Myrcene-based rubber
Tg	Filip Van Lijsebetten et al., 2022	−16 to −1°C	Modified VU networks
Tg	Lisa Sougrati et al., 2023	−28 to 18°C	Lignin-based vitrimers
Tg	W. Denissen et al., 2017	−25°C (elastomeric), 87°C (rigid)	Depending on formulation
Tg	Swagata Mondal et al., 2024	52 to 71°C	Doubly crosslinked networks
Tg	W. Denissen et al., 2018	~110°C	Vinylogous urea vitrimer
Damping Capacity			
Damping capacity	Jun Zhao et al., 2021	98%	Mechanically interlocked

The data reveal substantial variability in mechanical properties across different VU vitrimer systems. Young's modulus spans over six orders of magnitude, from approximately 31 kPa in soft PHU-based systems to 2.4 GPa in rigid bulk-polymerized networks. This range reflects the versatility of VU chemistry in accessing both elastomeric and thermosetting materials.

Stress Relaxation Behavior

Stress relaxation is a defining characteristic of vitrimers, and the relaxation times reported vary considerably based on temperature, composition, and catalytic additives.

Study	Relaxation Time	Temperature	Notes
Wim Denissen et al., 2015	85 s	170°C	Catalyst-free
J. Shin et al., 2022	~19 s	160°C	SIL-incorporated
P. Haida et al., 2020	0.3–54,000 s	110°C	Acid-catalyzed vs uncatalyzed
W. Denissen et al., 2017	2–10 min	120°C	With/without pTsOH catalyst
Lisa Sougrati et al., 2023	39–488 s	130°C	Lignin content dependent
Filip Van Lijsebetten et al., 2022a	20–4390 s	160°C	Depends on amine availability

The stress relaxation data demonstrate that relaxation times can be tuned over five orders of magnitude through compositional modifications and catalyst addition. The fastest relaxation times (~19 s at 160°C) were achieved through incorporation of solvate ionic liquids, which reduce the activation energy of transamination to 24 kJ/mol.

Factors Influencing Mechanical Properties

Chemical Composition Factors

Crosslinker Type and Concentration

The type and concentration of crosslinker significantly impact mechanical properties. Studies employing tris(2-aminoethyl)amine (TREN) as a trifunctional crosslinker generally produce networks with higher modulus compared to difunctional crosslinkers such as Priamine 1074. Increasing crosslinker content from 0.12 to 0.98 mol/L increased tensile strength from 0.18 to 1.68 MPa while decreasing elongation at break from 83% to 30%.

The relationship between crosslinking density (ν_e) and glass transition temperature follows classical theoretical models, with T_g positively correlated to ν_e . Similarly, the topology freezing temperature (T_v) is jointly determined by ν_e and the content of free amine groups.

Catalyst and Additive Effects

Catalytic additives profoundly influence both static and dynamic mechanical properties:

Additive Type	Effect on Relaxation	Effect on Modulus	Reference
pTsOH (Brønsted acid)	Decreases relaxation time by 5x	Minimal effect	W. Denissen et al., 2017
TBD (base)	Increases relaxation time	Improves dimensional stability	W. Denissen et al., 2017
Solvate ionic liquids	Decreases E_a to 24 kJ/mol	Maintains $>10^9$ Pa modulus	J. Shin et al., 2022
Zn(II) ions	Accelerates exchange	Significantly enhances via sacrificial bonds	L. Bai et al., 2020
LiTFSI salt	Decreases τ^* by ~70x	Decreases with plasticizing effect	Seongon Jang et al., 2023

Brønsted acids accelerate exchange reactions by protonating enamine-one species, creating electrophilic iminium-ones that facilitate transamination. Conversely, bases inhibit exchange reactions, with TBD incorporation increasing activation energy from 74 to 122 kJ/mol. This inhibitory effect improves compression set resistance, with only 5% permanent deformation observed.

Free Amine Content

The availability of free amine groups is critical for network dynamics. Networks with excess amine groups (VU-105) exhibit relaxation times of 20 s at 160°C, while those with excess acetoacetate (VU-95) show relaxation times of 4390 s under identical conditions. The strategic masking of primary amines through reversible trapping in dicarboxamide bonds enables temperature-dependent release of reactive chain ends, providing low creep at service temperatures while maintaining reprocessability at elevated temperatures.

Network Architecture Effects

Block Copolymer Structure

The architecture of block copolymer precursors significantly influences final vitrimer properties. Comparing statistical and diblock terpolymers of β -myrcene, styrene, and AAEMA, the diblock architecture (PS-*b*-P(Myrc-co-AAEMA)) exhibited four times higher tensile stress (3.1 ± 0.6 MPa) and nearly ten times higher hardness compared to statistical terpolymers. The addition of glassy PS domains anchored the network, reducing creep by 18–35%.

Block sequence also affects properties: BAB architectures produce hard materials, ABA produces tough materials, and AB produces elastic materials. This tunability arises from differences in how the crosslinkable B blocks interact with the glassy A blocks during network formation.

Mechanically Interlocked Networks

The incorporation of mechanically interlocked motifs provides enhanced mechanical performance through a consecutive energy dissipation mechanism. Vitrimers with slidable [2]rotaxane units showed Young's modulus of 18.5 ± 0.9 MPa versus 1.0 ± 0.1 MPa for non-slidable controls, toughness of 3.7 ± 0.1 versus 0.9 ± 0.1 MJ/m³, and damping capacity of 98% versus 72%. The intramolecular motions of rotaxanes accelerate dynamic exchange of VU bonds by loosening the network.

Dual Dynamic/Static Crosslinks

Incorporating both dynamic VU crosslinks and static crosslinks (e.g., from glycidyl methacrylate) creates networks with reduced creep. Addition of 5 mol% GMA decreased creep by 75% and imparted shape memory effects. Similarly, doubly crosslinked networks incorporating Janus-faced supramolecular moieties showed 41–46% less creep than single-network controls while conserving reprocessability.

Processing Condition Effects

Curing Temperature and Time

Standard curing protocols involve initial gelation at moderate temperatures (60–100°C) followed by post-curing at elevated temperatures (150°C). The kinetics of crosslinking tracked by in situ FTIR revealed that the apparent rate coefficient (k) is linearly correlated to crosslinking density (ν_e) when different polymer matrices are used.

Reprocessing Temperature

Reprocessing temperatures typically range from 100–180°C depending on the network composition. Higher temperatures enable faster stress relaxation but may risk thermal degradation. Optimal reprocessing for vinylogous urea composites was achieved at 180°C for 5 minutes, with no significant improvement at longer times or lower temperatures.

Bio-based and Sustainable Formulations

Several studies demonstrate that bio-based precursors can achieve competitive mechanical properties:

Bio-based Component	Mechanical Properties	Reference
Lignin (30–50 wt%)	E: 0.2–420 MPa, σ : 0.2–17.2 MPa	Lisa Sougrati et al., 2023
Castor oil/DL-limonene	Excellent reprocessability and shape memory	Yangyang Zhu et al., 2020
Cardanol-based	Successful fusion of distinct vitrimers	Fengbiao Chen et al., 2020
β -Myrcene	σ : 0.18–1.68 MPa, ϵ : 30–83%	Farhad Asempour et al., 2023
Starch (up to 70 wt%)	E: 2 GPa	P. Haida et al., 2023

Increasing lignin content increases tensile strength (from 0.2 to 17.2 MPa) and Young's modulus (from 0.2 to 420 MPa) but decreases elongation at break (from 98 to 22%). The vitrimer behavior allows stress-relaxation times as short as 39 s at 130°C with activation energies from 51 to 114 kJ/mol.

Effects of Reprocessing on Mechanical Properties

A key advantage of VU vitrimers is their ability to be reprocessed while retaining mechanical properties. The following table summarizes reprocessing studies:

Study	Cycles	Conditions	Property Retention	Notes
Wim Denissen et al., 2015	4	Grinding/compression molding	No significant degradation	Catalyst-free system
Mohammad Farkhondehnia et al., 2023	3	100°C, 15 metric tons, 45 min	Similar after 2 cycles, decrease after 3rd	PHU-based
Farhad Asempour et al., 2024	4	Hot pressing at 110°C	Minimal changes	Polymyrcene
C. Du et al., 2025	4	Hot pressing at 125°C	Thermomechanical properties retained	Block copolymer
Farhad Asempour et al., 2023	3	Grinding/hot pressing at 110°C	No appreciable changes	Myrcene rubber
Lisa Sougrati et al., 2023	3	Grinding/remolding at 140°C	71% stress at break after 1st cycle	Lignin-based
Swagata Mondal et al., 2024	5	Consistent protocol	Tg stable within $\pm 5^\circ\text{C}$	Doubly crosslinked
W. Denissen et al., 2018	2	180°C, 5 min compression	Similar properties after 2nd cycle	Vinylogous urea
Filip Van Lijsebetten et al., 2024	3	Shredding/remoulding, 150°C post-treatment	High retention, minimal degradation	Acid-catalyzed

The majority of studies report excellent property retention over multiple reprocessing cycles. However, some degradation mechanisms have been identified, including oxidation of free amine chain-ends at lower temperatures and

moisture introduction during reprocessing. Mechanical treatments such as grinding can cause initial property loss, though properties typically stabilize after the second cycle.

Synthesis

Reconciling Property Variability Across Studies

The substantial heterogeneity in reported mechanical properties reflects the fundamental versatility of VU chemistry rather than inconsistent findings. The six-order-of-magnitude range in Young's modulus (31 kPa to 2.4 GPa) can be understood through several key distinctions:

Polymer Matrix Effects: Elastomeric matrices such as PDMS, polytetrahydrofuran, and polyisoprene produce low-modulus materials (kPa to low MPa range), while rigid matrices based on phenoxy resins, epoxies, or aromatic structures yield high-modulus materials (GPa range). Studies using the same base chemistry but different matrices are not directly comparable.

Crosslinking Density Effects: Within a given matrix, crosslinking density systematically controls modulus and Tg. The linear correlation between the apparent rate coefficient (k) and crosslinking density (ν_e) provides a predictive framework for property design. Higher crosslinking density increases both modulus and Tg but reduces elongation at break.

Free Amine Availability: The apparent conflict between studies reporting fast relaxation (seconds) versus slow relaxation (hours) is resolved by considering free amine content. Networks designed with excess primary amines (VU-105 formulations) show relaxation times orders of magnitude faster than those with trapped or masked amines (VU-95 formulations). Both findings are correct within their respective design contexts.

Mechanistic Understanding of Catalyst Effects

The seemingly contradictory effects of different additives on mechanical properties can be reconciled through mechanistic analysis. Brønsted acids accelerate transamination by protonating the enamine-one to form an electrophilic iminium-one, while bases slow exchange by deprotonating reactive species. The thermoswitchable behavior of weak carboxylic acid catalysts—inhibiting at low temperatures through hydrogen bonding but promoting exchange at high temperatures through proton transfer—explains how the same additive can improve both service stability and reprocessability.

Lewis acidic metal ions (Zn^{2+} , Li^+) provide dual functionality: catalyzing exchange reactions while simultaneously forming sacrificial coordination bonds that enhance mechanical properties. The ~70-fold decrease in relaxation time upon lithium salt addition occurs alongside plasticization effects that reduce plateau modulus, representing a trade-off that can be optimized for specific applications.

Design Principles for Target Properties

Based on the collective evidence, the following design principles emerge:

For high modulus and strength: Use rigid aromatic or cycloaliphatic monomers, high crosslinking density, and reinforcing fillers (starch, silica, carbon nanotubes).

For fast reprocessing: Incorporate catalysts (acids, metal ions, ionic liquids), maintain free amine groups, and design for low activation energy of exchange.

For creep resistance: Mask primary amines in temperature-reversible groups, incorporate static crosslinks, add supramolecular reinforcement, or introduce phase-separated glassy domains.

For balanced properties: The PS-containing polyisoprene system (PS2IM2) exemplifies optimization, achieving both 100% toughness recovery during recycling and sensitive shape memory performance through the combination of dynamic VU bonds and phase-separated PS domains.

The steric factor (ρ) for VU bond exchange is critically important for viscosity predictions, with evidence suggesting values of $\rho < 10^{-10}$ are typical for vitrimers. This finding indicates that assumptions of $\rho = 1$ used in traditional analysis are invalid, and accurate property predictions require explicit consideration of steric effects alongside activation energy and free amine content.

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